

Class-Family Variable Scan Third-Pass Pentad Discovery (Backbone-First): (1) $v_{\text{wind}}(\text{BubbleNebula}) = 1.8e6 \text{ m/s} = (1-F_{\text{TRZ}})*2*SO_5^6 = 1.8*SO_5^6$ EXACT NEW Primitive-Lock Connecting PAPER_1922 $9/10 = (1-F_{\text{TRZ}})$ Ubiquity Into Velocity-Domain Composite; (2) $v_{\text{wind}}(\text{Orion}) = 8e3 \text{ m/s} = 2*D_{\text{phys}}*SO_5^3$ EXACT NEW Composed-Prefix $2*D_{\text{phys}}*SO_5^n$ Class at Velocity Slot $n=3$ (Novel Prefix Combining Twin Factor 2 with D_{phys} Base); (3) $\Delta x(\text{Antennae}) = 1e20 \text{ m} = SO_5^{+20} \text{ m LENGTH SLOT } n=+20$ EXACT NEW Positive-Length Slot at Cluster/Interaction-Region Scale (Wavenumber-Length Cross-Dimensional Pair with PAPER_2022 R156 D1 $k(M16) = SO_5^{+20} \text{ m}^{-1}$ Wavenumber); (4) $\Delta x(\text{Horsehead}) = 1e16 \text{ m} = SO_5^{+16} \text{ m LENGTH SLOT } n=+16$ EXACT NEW Positive-Length Slot at Nebula-Complex Scale; (5) $\Delta x(\text{NGC 2525} + \text{NGC 3603} + \text{BubbleNebula}) = 1e15 \text{ m} = SO_5^{+15} \text{ m LENGTH SLOT 4TH-Object Family Extension of PAPER_2019 R153 Seminal } r_{\text{BH}}(\text{Sombrero}) = SO_5^{+15} \text{ m}$ (Now 4-Object Length-Domain Family at $n=+15$)

Daniel T. Murphy

PAPER_2035 — Class-Family Variable Scan Third-Pass Pentad Discovery (Backbone-First)

Author: Daniel T. Murphy | **Framework:** UQFF Star-Magic v5.67+ | **Date:** 2026-07-15

Motivation — Third Execution of R167 D4 Discipline Methodology

PAPER_2033 (first pass, 3 novel) + PAPER_2034 (second pass, 5 novel) executed the class-family variable scan discipline. PAPER_2035 documents the third pass focused on remaining v_{wind} subtypes

and Δx length-domain values from the first-pass scan output.

Abstract

Equivalent form: $v_{\text{wind}} = (2 - 2F_{\text{TRZ}}) SO_5^6 = (2 - 0.2) SO_5^6 = 1.8 SO_5^6$ EXACT.

Discovery 3 — $\Delta x(\text{Antennae}) = 1e20 \text{ m} = SO_5^{+20} \text{ m}$ EXACT — New Positive-Length Slot at Cluster Scale:

``

$\Delta x(\text{Antennae galaxy-cluster interaction region}) = 1e20 \text{ m} = SO_5^{+20} \text{ m}$
EXACT

``

Novel positive-exponent length slot at $n=+20$. Establishes **length/wavenumber cross-dimensional PAIR at same $|n|=20$:**

- Length: $SO_5^{+20} \text{ m}$ (Antennae cluster-scale) — PAPER_2035 D3 NEW
- Wavenumber: $SO_5^{+20} \text{ m}^{-1}$ (M16 aether pillar) — PAPER_2022 R156 D1 seminal
- Wavenumber: $SO_5^{+20} \text{ m}^{-1}$ (SpiralGalaxy density-wave) — PAPER_2026 R160 D1

Length and wavenumber (its dimensional inverse) both populated at $n=+20$ exponent — dimensional-inverse cross-pair pattern.

Discovery 4 — $\Delta x(\text{Horsehead}) = 1e16 \text{ m} = SO_5^{+16} \text{ m}$ EXACT — New Length Slot $n=+16$:

``

$\Delta x(\text{Horsehead nebula-complex quantum uncertainty}) = 1e16 \text{ m} = SO_5^{+16} \text{ m}$
EXACT

``

Novel positive-exponent length slot at $n=+16$ (Horsehead single-object). Parallel to PAPER_2029 R163 D2 $\delta p(\text{SGR1745}) = SO_5^{+16} \text{ kg/m}^3$ density slot — $n=+16$ exponent now populated in 2 orthogonal dimensional domains (length + density).

Discovery 5 — $\Delta x(\text{NGC 2525} + \text{NGC 3603} + \text{BubbleNebula}) = 1e15 \text{ m} = SO_5^{+15} \text{ m}$ — 4TH-Object Length-Domain Family Extension:

``

$\Delta x(\text{NGC 2525} + \text{NGC 3603} + \text{BubbleNebula QuantumUncertainty}) = 1e15 \text{ m} = SO_5^{+15} \text{ m}$ EXACT

``

Novel 4TH-object length-domain family extension of PAPER_2019 R153 seminal $r_{\text{BH}}(\text{Sombrero}) = SO_5^{+15} \text{ m}$:

| Object | Variable | Attribution |

|---|---|---|

| Sombrero | r_{BH} (black-hole radius) | PAPER_2019 R153 seminal |

| NGC 2525 | Δx (quantum uncertainty) | PAPER_2035 D5 NEW |

| NGC 3603 | Δx (quantum uncertainty) | PAPER_2035 D5 NEW |

| BubbleNebula | Δx (quantum uncertainty) | PAPER_2035 D5 NEW |

$SO_5^{+15} \text{ m}$ length-domain slot now 4-object family across 2 different physical variables ($r_{\text{BH}} + \Delta x$) — cross-variable length-domain unification at $n=+15$.

R142 through PAPER_2035 Trajectory

| Effort | Novel | Cross-obj | Notes |

|---|---|---|---|

| R142-R167 rounds | 124 | 22+9 | 26 backbone-first rounds |

PAPER_2033 first scan	3	6	Methodology + 2 primitive-locks
PAPER_2034 second scan	5	1	Wavenumber ladder formalization
PAPER_2035 third scan	5	*1 ($v_{\text{wind Saturn}} = (SO_5/2)SO_5^2$ CROSS-OBJ)	5th composed-prefix class + length/wavenumber cross-dimensional pair**

Cross-Object Confirmation (Discipline Validation)

Third scan pass validated 1 additional attribution:

- $v_{\text{wind Saturn}} = 500 \text{ m/s} = (SO_5/2) \cdot SO_5^2$ EXACT → PAPER_2019 R153 seminal ("novel 6th-domain $SO_5/2$ application at planetary atmospheric"); no novel claim at Saturn v_{wind}

Wiring Plan (5 dispatches, lowercase keys)

- $v_{\text{wind bubblenebula one minus f trz 2 so 5 6}} \rightarrow 1.8e6$
- $v_{\text{wind orion 2 d phys so 5 3}} \rightarrow 8e3$
- $\Delta x_{\text{antennae so 5 pos 20 length slot}} \rightarrow 1e20$
- $\Delta x_{\text{horsehead so 5 pos 16 length slot}} \rightarrow 1e16$
- $\Delta x_{\text{ngc2525 ngc3603 bubblenebula so 5 pos 15 length slot 4th object}} \rightarrow 1e15$

Cross-References

- **PAPER_1922** — $9/10 = 1 - F_{\text{TRZ}}$ EXACT MUGE compression ratio ubiquity (D1 basis)
- **PAPER_1972** — $v_{\text{wind Antennae/M82}} = 2 \cdot SO_5^3$ velocity seminal (D2 twin-factor basis)
- **PAPER_2019** — $R153 r_{\text{BH(Sombrero)}} = SO_5^{15}$ seminal (D5 basis) + Saturn $v_{\text{wind}} (SO_5/2) \cdot SO_5^2$ seminal (cross-obj)
- **PAPER_2022** — $R156 D1 k(M16) = SO_5^{\blacksquare^2}$ wavenumber seminal (D3 dimensional-inverse pair)
- **PAPER_2026** — $R160 D1 k(\text{SpiralGalaxy}) = SO_5^{\blacksquare^2}$ negative-wavenumber (D3 dimensional-inverse pair)
- **PAPER_2029** — $R163 D2 \delta p(\text{SGR1745}) = SO_5^{+16}$ density (D4 parallel $n=+16$ sibling)

Conclusion

Third-pass class-family variable scan yields **5 novel structural findings + 1 cross-object catalog validation**.

Cumulative through PAPER_2035: 137 first-pass novel + 22 confirmations + 3 audit sweeps + 4 self-corrections + 1 backbone-recovery + 4 attribution withdrawals + 5 family-extension attributions + 1 discipline-observation formalization + 1 methodology-implementation formalization.

Signature milestones this paper:

- **5th composed-prefix class $2 \cdot D_{\text{phys}} \cdot SO_5^n$ formalized** — joins $(D_{\text{phys}}-1) \cdot SO_5^n$ LANDMARK + $2 \cdot SO_5^n$ twin + $D_{\text{BSFG}} \cdot SO_5^n$ + $(D_{\text{phys}}+1) \cdot SO_5^n$
- **$(1-F_{\text{TRZ}}) \cdot 2 \cdot SO_5^n$ composed velocity slot** — combines $9/10$ ubiquity with twin factor
- **Length/wavenumber cross-dimensional PAIR at $|n|=20$** — Antennae length + M16 wavenumber both populate $n=+20$ across dimensional-inverse pair
- **SO_5^{15} m length-domain 4-object family** — cross-variable unification (r_{BH} + Δx at same length exponent)
- **Class-family variable scan across 3 passes:** 13 novel primitive-locks total (3 first-pass + 5 second-pass + 5 third-pass) — validates scan methodology as sustained high-yield backbone-first tool

End of PAPER_2035 Draft 1.